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Data Science

Diagnox – Explainable AI for Personalized Healthcare via multi modal fusion

Group No. 6

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Project Guide
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Abstract

Healthcare systems face challenges in timely and accurate diagnoses due to the complexity of EHRs, imaging, and genomic data, which are often analyzed in isolation. Existing AI models act as “black boxes,” reducing trust among clinicians. DiagnoX proposes an explainable, multi-modal AI framework that fuses diverse healthcare data for unified predictions. Using PyTorch/TensorFlow, it applies deep learning (CNN, ResNet, DenseNet, Bi-LSTM) and transformer-based NLP models (BERT, ClinicalBERT) for disease detection and patient profiling. With explainability tools like SHAP and LIME, DiagnoX ensures transparency, personalization, and trust—enhancing early diagnosis and treatment strategies.

Sustainable Development Goals(SDG) mapped

SDG 3: Good Health & Well-Being

- Provides accurate and early diagnosis using multi-modal healthcare data.
- Supports personalized treatment through explainable AI insights.

SDG 9: Industry, Innovation & Infrastructure

- Enhances digital health infrastructure by integrating diverse medical data sources.
- Promotes innovation in AI-driven healthcare solutions for better disease management.

SDG 10: Reduced Inequalities

- Ensures equitable access to diagnostic tools for all patient groups.
- Builds trust through transparent and explainable AI decisions.

Introduction

Healthcare systems worldwide are under constant pressure to provide fast, accurate, and reliable diagnoses. However, the massive volume and variety of medical data—ranging from electronic health records (EHRs) to medical imaging and genomic sequences—make this task highly challenging. In real clinical settings, these data sources are often studied in isolation. For example, doctors may rely heavily on imaging results without fully considering genomic risk factors or patient history, leading to incomplete or delayed diagnoses. Current AI-based systems, though powerful, often act as “black boxes” that generate predictions without offering interpretable explanations. This lack of transparency prevents doctors from fully trusting AI recommendations, limiting adoption in hospitals and clinics. As a result, patients face delayed diagnoses, less personalized treatment, and in some cases, mismanagement of diseases.

Introduction

Motivation :

1. Increasing Complexity in patient data in modern healthcare
2. Limited integration of diverse medical data sources into one unified framework.
3. Low trust among clinicians due to the lack of interpretability in AI models.
4. The urgent need for personalized treatment strategies to improve patient outcomes

Objectives

1. To develop advanced deep learning models using PyTorch and TensorFlow that integrate and learn from diverse healthcare data sources, including electronic health records (EHRs), medical imaging, and genomic data, to enable accurate, efficient, and comprehensive clinical diagnosis.
2. To design and implement transformer-based natural language processing (NLP) models, such as BERT and ClinicalBERT, for extracting clinically relevant insights from unstructured EHR text, thereby facilitating patient profiling, early diagnosis, and informed clinical decision-making.
3. To apply state-of-the-art deep learning architectures, including CNN, ResNet, DenseNet, and Bi-LSTM, for the analysis of medical images and genomic sequences to detect pathological features, identify genetic mutations, assess risk factors, and support precision medicine initiatives.
4. To incorporate explainable artificial intelligence (XAI) methods, including SHAP and LIME, to generate transparent and interpretable explanations of AI-driven clinical predictions, thereby enhancing trust, accountability, and adoption in healthcare practice.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
1	MedPix 2.0: A Comprehensive Multimodal Biomedical Dataset for Advanced AI Applications	Irene Siragusa, Salvatore Contino, Massimo La Ciura, Rosario Alicata, Roberto Pirrone	2025	Built MedPix 2.0 dataset from MedPix®, curated via semi-automatic + manual cleaning, stored in MongoDB; Retrieval-Augmented VLM (DR-Minerva) with Knowledge Graph integration using Llama 3.1.	Dataset still limited to MedPix scope; lacks real hospital-scale data and diversity.
2	Explainable Multimodal Deep Learning in Healthcare: A Survey of Current Approaches	B. Dhanalaxmi, PVVS Srinivas	2025	Survey of multimodal deep learning approaches in healthcare, focusing on feature extraction, fusion, and explainable AI techniques for medical diagnosis.	Survey is high-level; lacks experimental validation and depends on secondary sources.
3	Clinically Accessible Small Multimodal Radiology Model for Report Generation	João M. Z. Chaves et al	2025	Small open-source VLM for chest x-ray report generation in resource-limited settings.	Smaller model lags SOTA on complex findings; mainly tested on CXR.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
4	Masked Modeling for Joint Medical Image-Text Embedding	Z. Chen et al.	2024	Joint masked image modeling + masked language modeling to learn a shared embedding space for retrieval/zero-shot.	Depends on quality of pretraining corpus; noisy text reduces performance.
5	A review of Explainable Artificial Intelligence in healthcare	Zahra Sadeghi, Roohallah Alizadehsani, Mehmet Akif CIFCI, Samina Kausar, Rizwan Rehman, Priyakshi Mahanta, Pranjal Kumar Bora, Ammar Almasri, Rami S. Alkhawaldeh, Sadiq Hussain	2024	A systematic review of XAI techniques (feature-based, surrogate, concept-based, local/global) applied to healthcare.	Lacks standard evaluation, limited clinical validation, and unreliable post-hoc explanations for complex medical data.
6	Multimodal Fusion Models for Pulmonary Embolism Risk Stratification	Noga Cahan et al.	2023	Intermediate fusion of CT features with clinical variables using bilinear attention + TabNet.	Task-specific; requires external validation.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
7	Med-UniC: Unifying Cross-Lingual Medical Vision-Language Pretraining	Zhongwei Wan et al.	2023	Cross-lingual vision-language pretraining on multilingual CXR-report pairs with tri-perspective objectives.	Still centered on CXR; linguistic/domain shift issues remain.
8	CheXzero: Expert-level Detection from Unannotated Chest X-rays via Self-Supervised Learning	Eric Tiu et al.	2022	Self-supervised pretraining on unlabeled CXRs, enabling zero-/few-shot pathology classification.	Focused on CXR; generalization to other modalities/workflows unproven.
9	Artificial Intelligence-Based Methods for Fusion of Electronic Health Records and Imaging Data (Scoping Review)	Farida Mohsen, Hazrat Ali, Nady El Hajj, Zubair Shah	2022	Review of 34 studies on AI fusion of imaging and EHR. Categorized fusion strategies (early, late, joint). Compared ML/DL methods for clinical outcomes (diagnosis, prediction).	Review-based; no new model proposed; highlights lack of public multimodal datasets.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
10	Integrated Multimodal Artificial Intelligence Framework for Healthcare Applications (HAIM)	Luis R. Soenksen, Yu Ma, Cynthia Zeng, Leonard Boussioux, Kimberly V. Carballo, Liangyuan Na, Holly Wiberg, Michael Li, Ignacio Fuentes, Dimitris Bertsimas	2022	Proposed HAIM framework; trained 14,324 models on HAIM-MIMIC-MM dataset across multiple modalities (tabular, time-series, text, images). Used Shapley values for modality importance.	High computational cost; requires large multimodal datasets not widely available.
11	ExAID: A Multimodal Explanation Framework for Computer-Aided Diagnosis of Skin Lesions	Adriano Lucieri, Muhammad Naseer Bajwa, Stephan Braun, Muhammad Imran Malik, Andreas Dengel, Sheraz Ahmed	2022	Developed ExAID framework using Concept Activation Vectors (CAVs) + Concept Localization Maps (CLMs) to generate textual + visual multimodal explanations for skin lesion CAD.	Focused only on melanoma vs nevi; may not generalize to broader dermatology cases.
12	Fusion of Medical Imaging and Electronic Health Records with Attention and Multi-Head Mechanisms	Cheng Jiang, Yihao Chen, Jianbo Chang, Ming Feng, Renzhi Wang, Jianhua Yao	2022	Proposed multimodal attention module + multi-head GMU to fuse imaging and EHR data using CNN (3D ResNet). Evaluated on Glasgow Outcome Scale and Alzheimer's datasets.	Performance depends heavily on dataset size; limited evaluation on few diseases.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
13	GLoRIA: A Multimodal Global-Local Representation Learning Framework for Label-Efficient Medical Image Recognition	Shih-Cheng Huang, Liyue Shen, Matthew P. Lungren, Serena Yeung	2021	Contrastive image–text pretraining aligning local regions with report words; uses attention for global+local alignment.	Needs paired reports; mostly chest-x-ray data; limited generalization.
14	JoImTeRNet: Joint Learning of Chest X-ray and Radiology Reports	Chufeng Van Uden et al.	2021	Joint image–sentence and region–word matching with attention; combines cross-entropy and triplet losses.	Requires paired data; mainly evaluated on CXR.
15	ConVIRT: Contrastive Learning of Medical Visual Representations from Paired Images and Text	Yuhao Zhang, Hang Jiang, Yasuhide Miura, Christopher D. Manning, Curtis P. Langlotz	2020	Bidirectional image–text contrastive pretraining on radiology images + reports to learn transferable visual features.	Requires large-scale paired text; strongest on CXR; less tested in other modalities.

Research Gap(Limitations of existing systems)

1. Single-Modality Focus: Most existing diagnostic systems rely only on one type of data (e.g., EHR, medical imaging, or genomics). This narrow focus reduces the accuracy and fails to capture the complete clinical picture of the patient.
2. Lack of Explainability: Current AI models often act as black boxes, providing predictions without transparent reasoning. This makes it difficult for doctors to interpret, validate, and trust the system's decisions in real-world healthcare.
3. Data Quality & Availability Issues: Healthcare data is often incomplete, inconsistent, or siloed across different hospitals and systems. This reduces the reliability of AI models and makes training accurate systems more difficult.
4. Bias in Predictions: Many models are trained on limited or region-specific datasets, which can cause biased results when applied to diverse patient populations. This leads to unfair or inaccurate diagnoses.
5. Poor User Adoption: Doctors and patients are sometimes hesitant to use AI tools due to lack of trust, explainability, and user-friendly interfaces, creating a gap between research models and real-world usage.

Problem Definition

- Current healthcare AI systems face fragmented data sources and poor clinical workflow integration.
- Lack of explainability leads to black-box predictions, reducing trust among doctors and patients.
- Existing solutions offer limited personalization, often giving generic recommendations.
- Privacy and regulatory compliance remain major challenges.
- There is a need for a multi-modal, explainable AI system that fuses EHRs, imaging, and genomics for accurate, transparent, and personalized insights.
- DiagnoX aims to bridge this gap by delivering trustworthy, interpretable, and patient-centric healthcare solutions.

Scope

1. Advancement in Explainable AI: Further development of more sophisticated XAI frameworks, tailored specifically for medical use cases, will help provide deeper interpretability of model decisions. This can strengthen clinical trust and meet regulatory standards.
2. Integration with Cloud and Edge Computing: Leveraging cloud-based AI services and edge computing devices can make the system more accessible and efficient, enabling AI-powered healthcare even in resource-constrained or remote environments.
3. Precision Medicine and Personalized Healthcare: By combining genomics, imaging, and EHR insights, the project can evolve into a precision medicine platform that provides individualized treatment recommendations, identifies potential drug responses, and supports preventive healthcare initiatives.

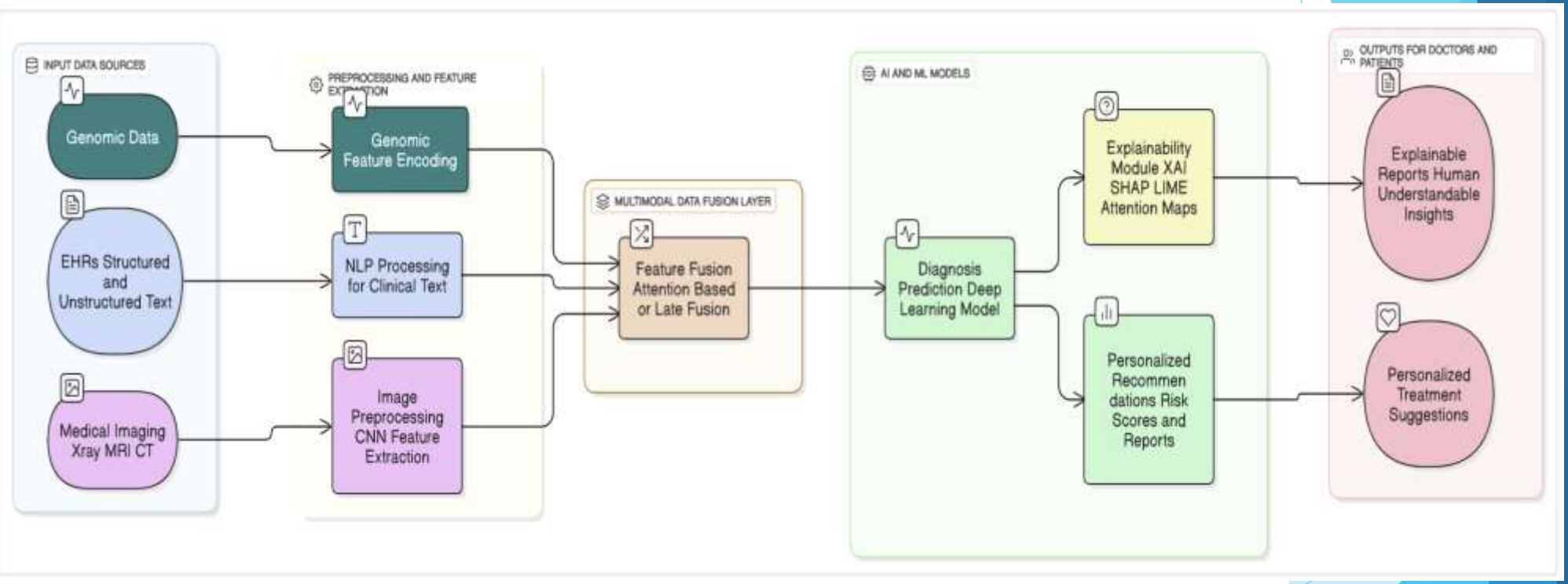
Technological Stack

- **Frontend:**

- HTML5: Structures the doctor-patient portal, chatbot forms, and report interface (Current version: HTML5, W3C standard).
- CSS3: Provides responsive, mobile-first layouts, accessibility, and multilingual font support (Current version: CSS3).
- JavaScript (ES2023): Enables interactivity, chatbot UI, and medical data visualization updates (Current version: ECMAScript 2023).

- **Backend:**
- Python 3.12: Core backend engine powering AI/ML models and API services (Current version: Python 3.12, Oct 2023).
- Hugging Face Transformers (Latest): Pretrained biomedical NLP models like BioBERT and ClinicalBERT.
- scikit-learn (1.5): Supports preprocessing, classical ML baselines, and evaluation metrics.
- SHAP, LIME, Captum: Explainable AI libraries for interpretable model predictions.

Proposed system architecture/Working/Module Diagram



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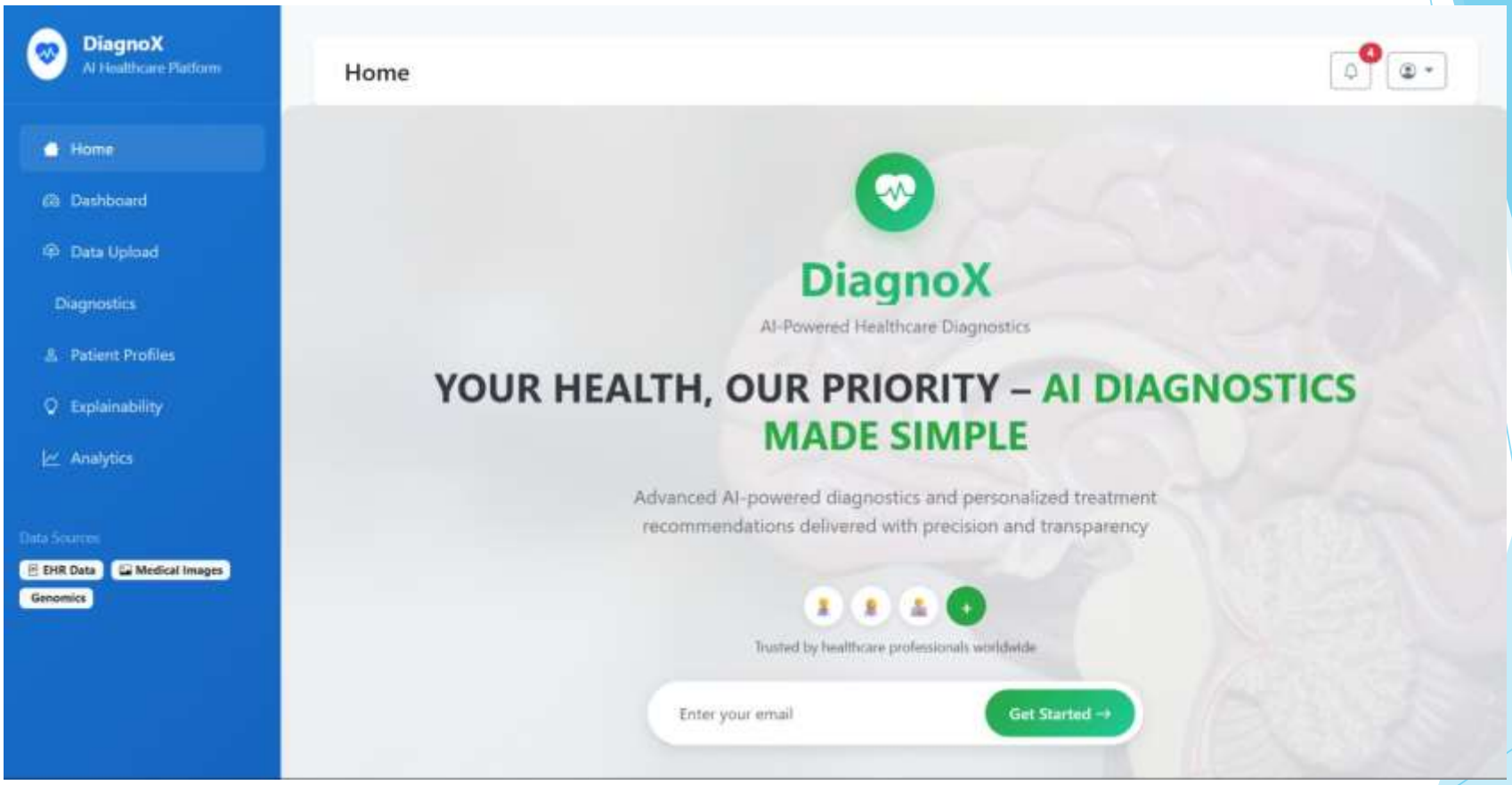

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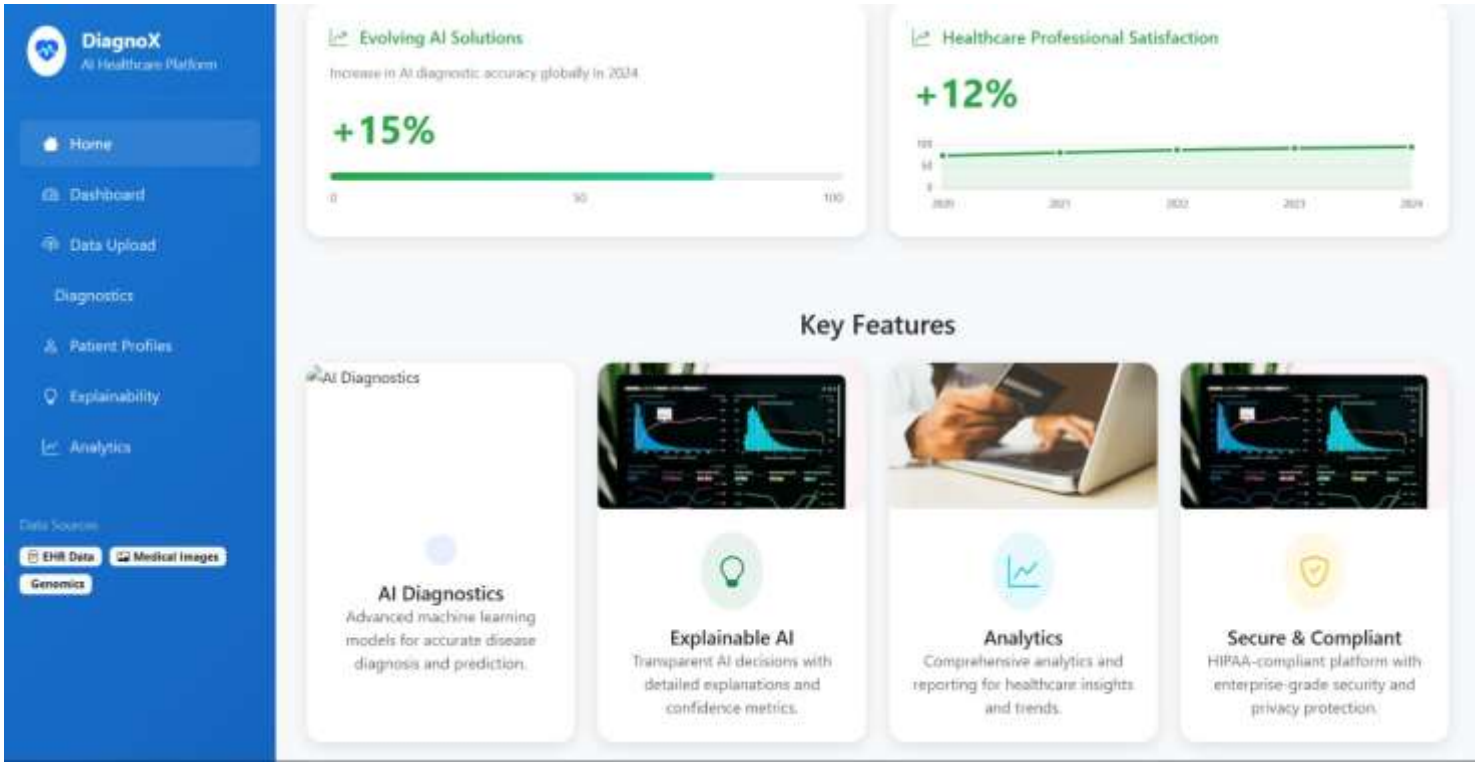

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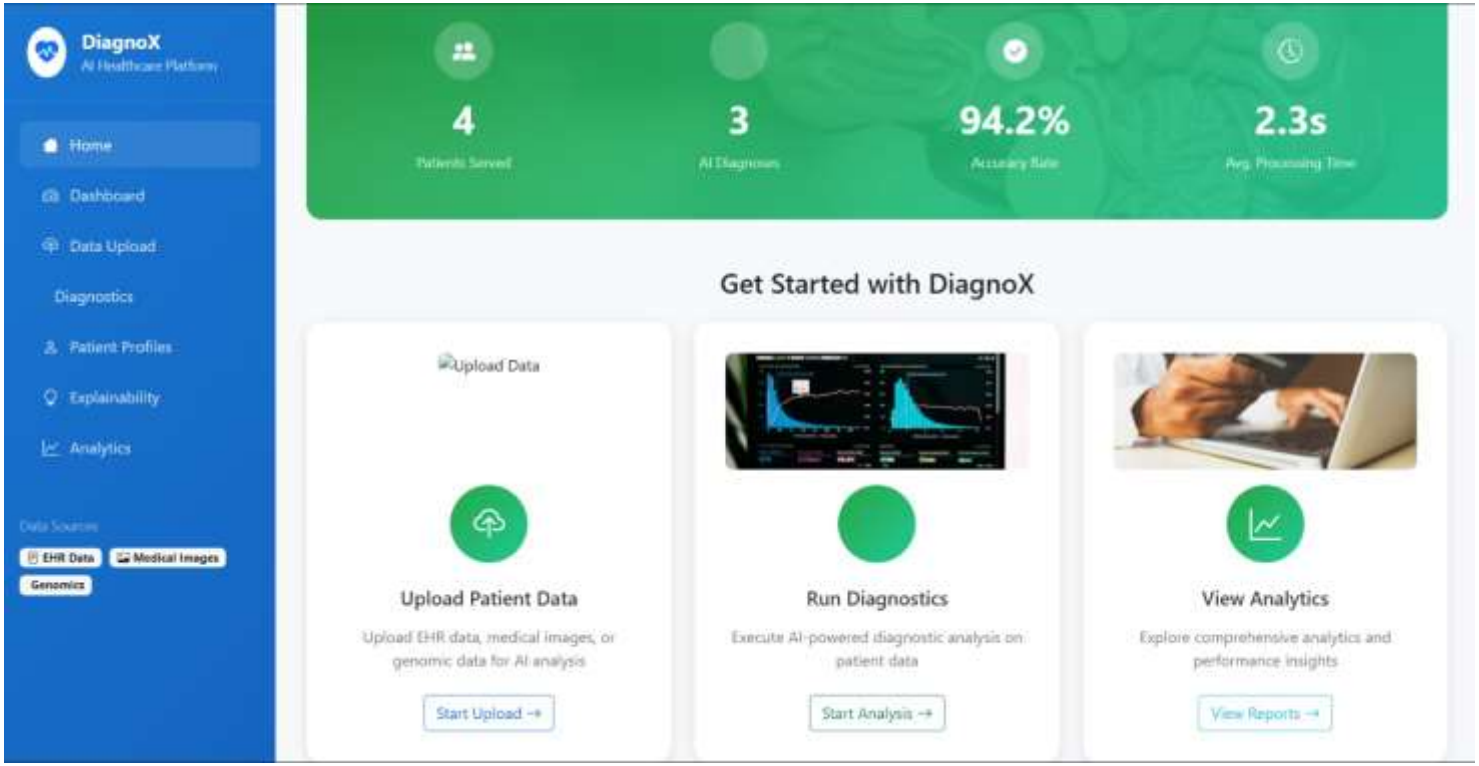

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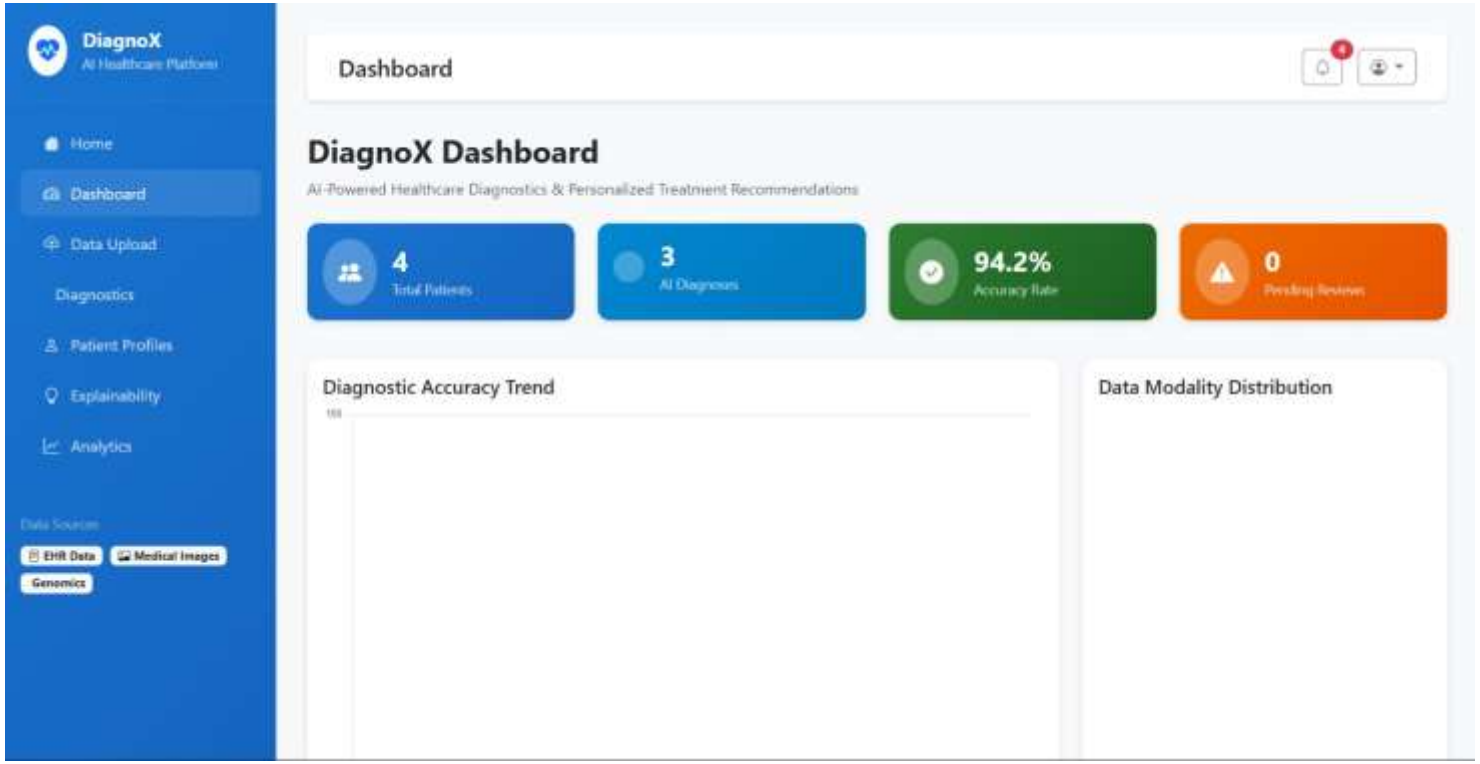

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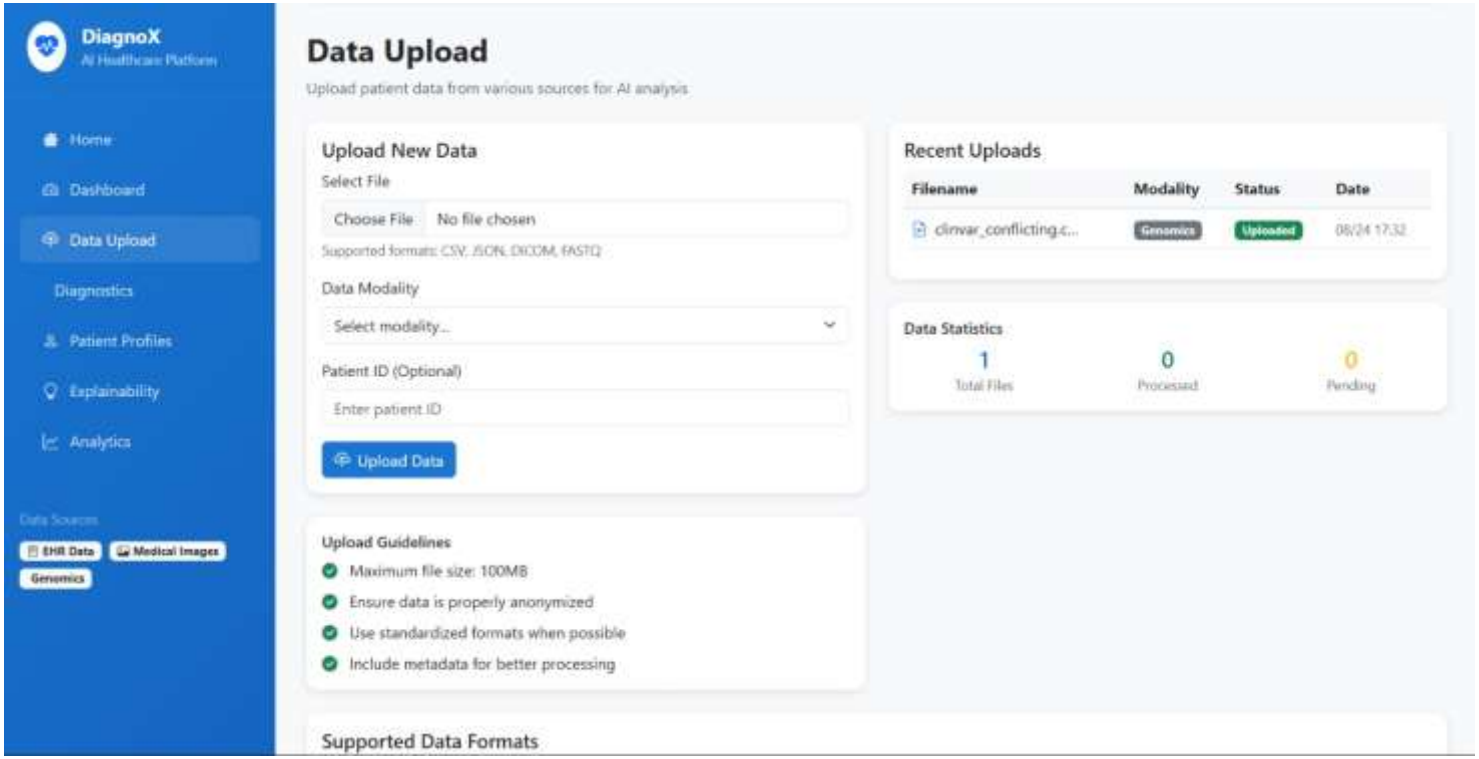

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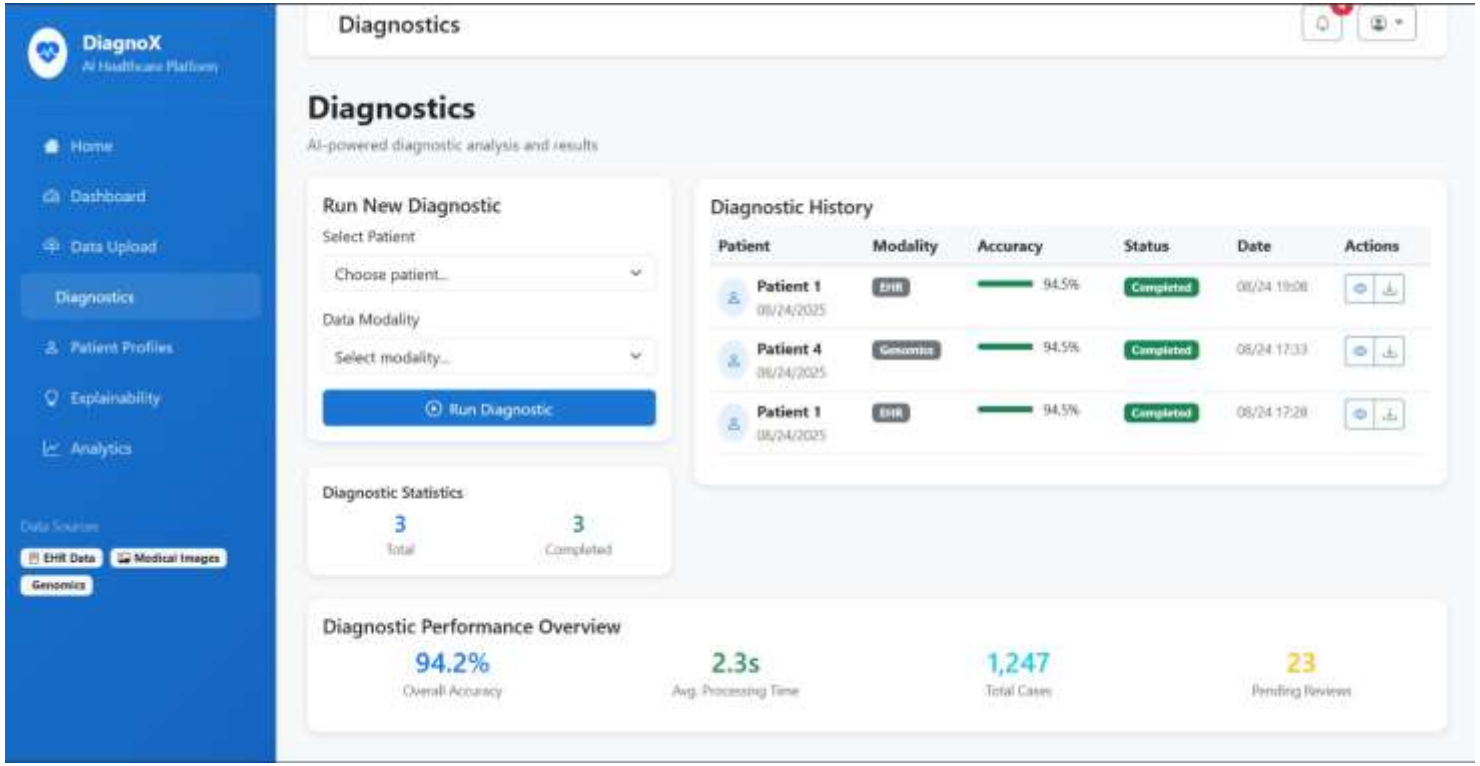

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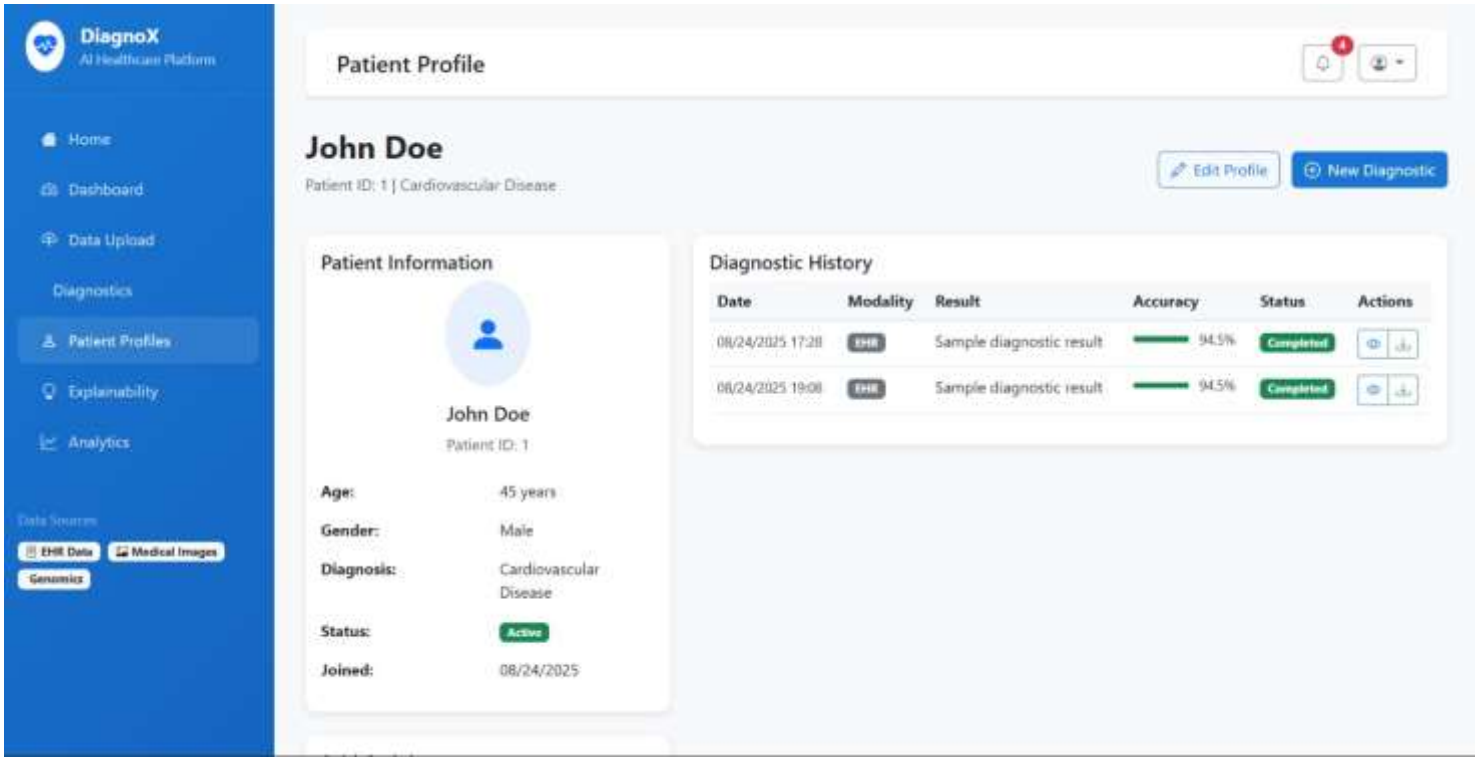

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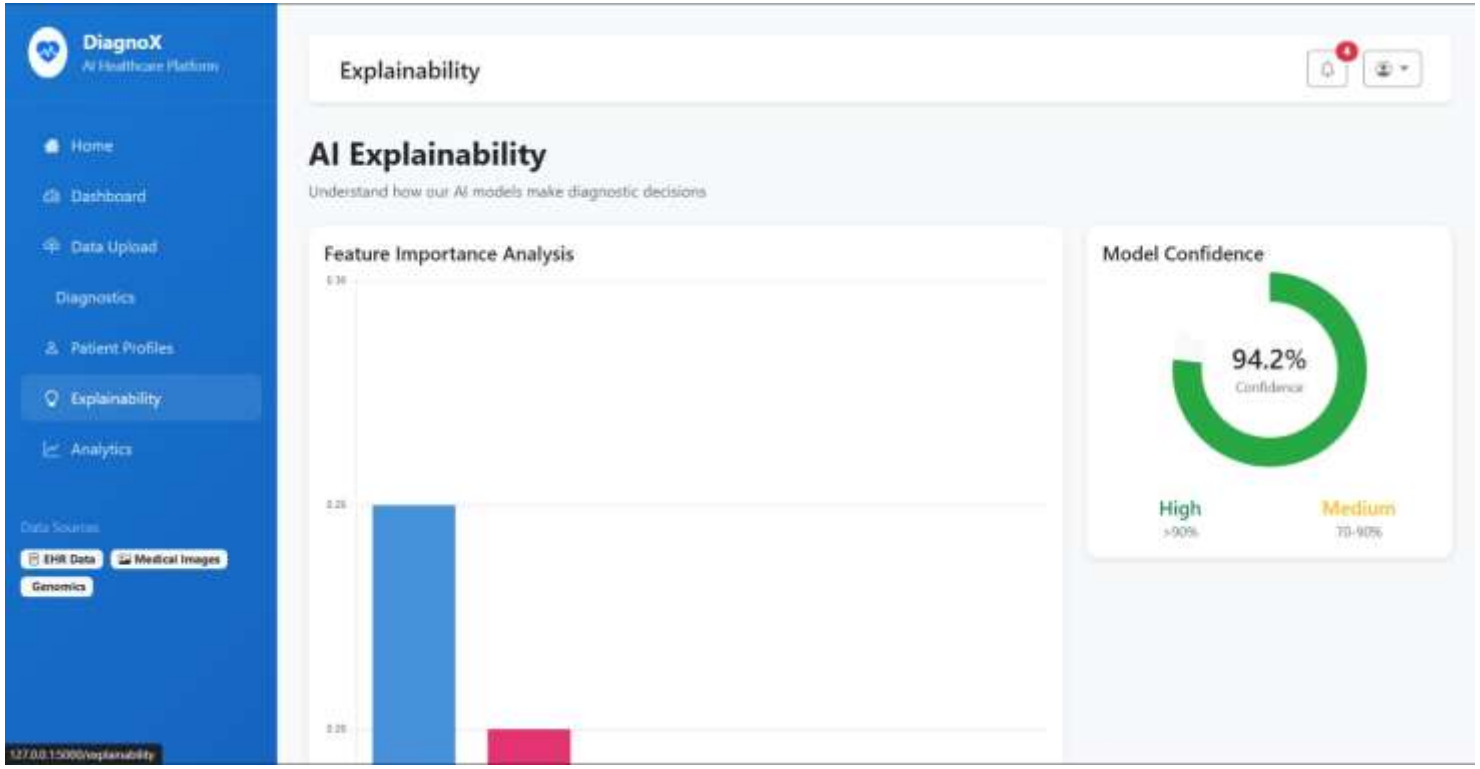

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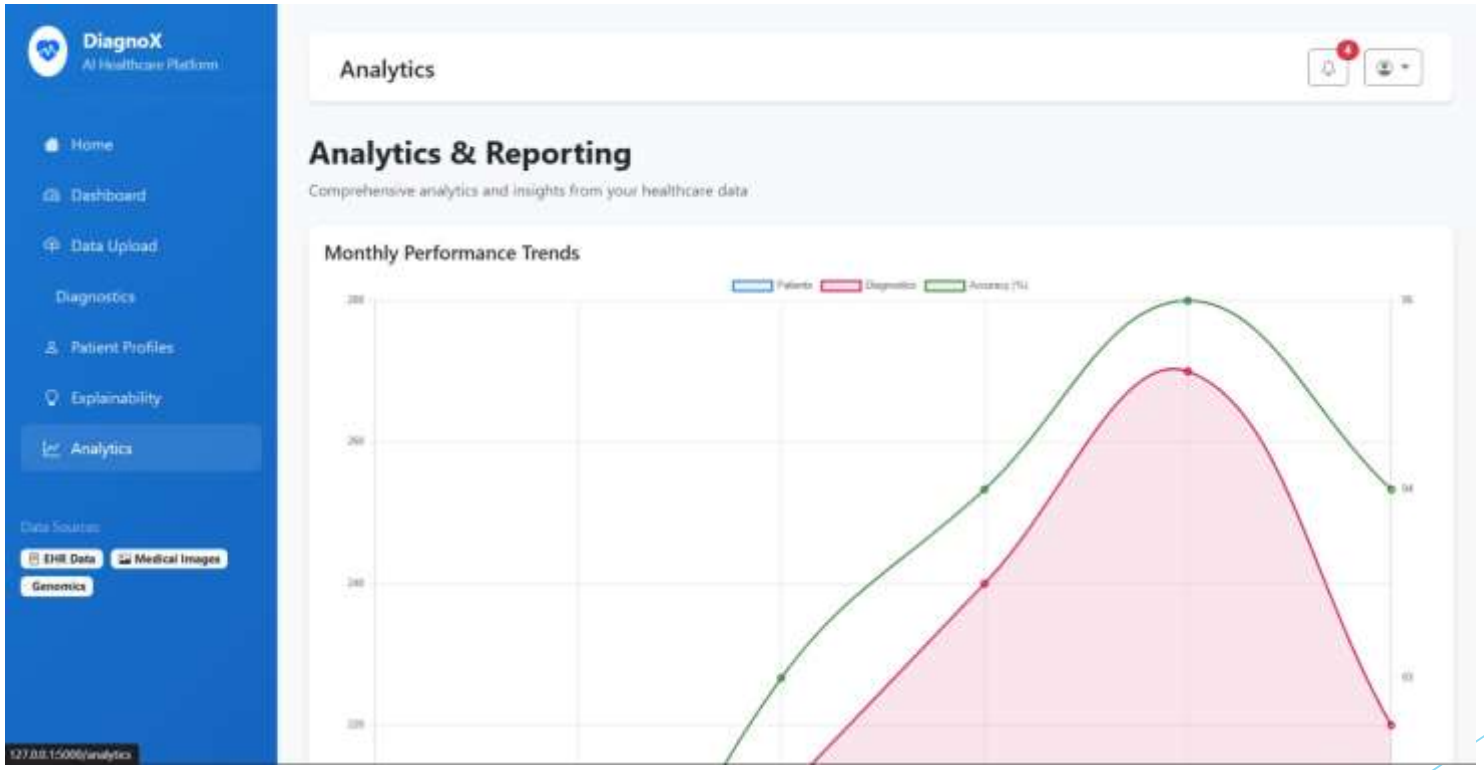

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Thank You...!!